YOUSEF ELBEYALY

Machine Learning & Generative AI Engineer

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Availability: Open for Remote / Hybrid Roles
Languages: Arabic, English

PROFESSIONAL SUMMARY

Results-oriented AI/ML & Deep Learning Engineer with a rigorous foundation in mathematical modeling, computational statistics, and software engineering. Proven track record in architecting end-to-end Generative AI & NLP applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) frameworks, and Vector Databases, as well as developing high-performance Computer Vision pipelines (PyTorch, OpenCV). Expert in bridging complex theoretical AI architectures—including Quantization (FP16/BFLOAT16) and Transformer decoding—into scalable, production-ready microservices and interactive deployments.

TECHNICAL SKILLS & COMPETENCIES

Generative AI & LLMs	Hugging Face Transformers, Large Language Models (LLaMA, T5, FLAN-T5), RAG Architectures, Sentence-Transformers, Prompt Engineering, Model Quantization (FP16, BFLOAT16), Fine-tuning (LoRA/PEFT), LangChain, Agentic Workflows, NLP Pipelines.
Databases & Vector Search	Qdrant, pgvector, FAISS (CPU), ChromaDB, SQL, Vector Embeddings, Semantic Similarity Search, Cosine Similarity Indexing.
Deployment & MLOps	Docker & Docker Compose, Celery, RabbitMQ, Redis, Nginx, Prometheus, Grafana, Streamlit, FastAPI, Uvicorn, Git, Amazon Web Services (AWS), Hugging Face Spaces, Linux/WSL Environments.
Machine & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, CatBoost, Optuna, Computational Statistics, Time-Series Forecasting, Transfer Learning (VGG16, ResNet).
Computer Vision & NLP	OpenCV, MediaPipe, NLTK, pdfplumber, Document Processing & Text Extraction, Hand Landmark Detection, Image Classification & Segmentation.
Programming Languages	Python (Advanced, OOP, Modular Packaging), SQL, C#.

PROFESSIONAL EXPERIENCE

Freelance Machine Learning & AI Engineer **Jan 2026 – Present**

- Architected and deployed end-to-end Machine Learning and Generative AI solutions, integrating custom Deep Learning pipelines and LLM inference engines via production-ready RESTful APIs (FastAPI, Uvicorn).
- Engineered Retrieval-Augmented Generation (RAG) document assistants, utilizing vector databases (FAISS, Qdrant) and sentence embeddings to enable high-accuracy semantic search and automated text summarization.
- Researched and implemented advanced theoretical deep learning architectures, including Attention mechanisms, Flow Matching, and Diffusion Models, optimizing local model performance and inference speed.
- Consulted on data leakage prevention, feature engineering, and hyperparameter tuning (Optuna) for tabular revenue forecasting and regression projects.

FEATURED ENGINEERING PROJECTS

Distributed Mini-RAG & Microservices Architecture **2026**

Tech Stack: *FastAPI, Celery, RabbitMQ, Redis, Qdrant, pgvector, Docker Compose, Nginx, Prometheus, Grafana*

- Architected an asynchronous, production-grade RAG backend separating heavy document processing and vector indexing into dedicated Celery worker queues powered by RabbitMQ and Redis.

- Integrated hybrid vector search using Qdrant and pgvector (PostgreSQL 17) to achieve low-latency document retrieval and context construction without blocking the FastAPI event loop.
- Deployed a complete multi-container environment via **Docker Compose** with **Nginx** reverse proxying and real-time observability using **Prometheus, Grafana, Node-Exporter, and Flower task monitoring**.

PDF AI Assistant (End-to-End RAG & LLM System) **2026**

Tech Stack: *Hugging Face Transformers (T5/FLAN-T5), FAISS, Sentence-Transformers, pdfplumber, Streamlit*

- Designed and deployed a local Retrieval-Augmented Generation (RAG) assistant capable of ingesting multi-page PDF reports, extracting structured text, and performing context-aware Q&A and automated chapter summarization.
- Built a high-performance semantic search pipeline using sentence-transformers and FAISS CPU index to retrieve top-k relevant document chunks in milliseconds with zero data leakage.
- Optimized local LLM decoding and inference, applying memory-efficient techniques and constructing an interactive, professional Web UI using Streamlit for seamless user ingestion and real-time chat.

Chest Cancer Detection & X-Ray Classification System **2025**

Tech Stack: *Tensor Flow, Transfer Learning (VGG16), OpenCV*

- Developed a high-precision Computer Vision diagnostic tool for classifying chest X-ray scans using Tensor Flow and Transfer Learning with deep convolutional networks (VGG16 architecture).
- Implemented advanced image preprocessing, normalization, and data augmentation techniques to prevent model overfitting and significantly enhance classification recall and F1-score on medical datasets.

AI Virtual Mouse & Real-Time Fitness Exercise Trainer **2025**

Tech Stack: *Python, OpenCV, MediaPipe, Real-Time HCI Processing*

- Engineered a low-latency Human-Computer Interaction (HCI) system utilizing MediaPipe hand and body landmark detection to enable gesture-based mouse control without physical hardware.
- Built an automated fitness tracking module that calculates joint angles in real-time from webcam feeds, accurately counting repetitions and evaluating posture exercise mechanics.

AI Mizan - Intelligent Food Price Forecasting Engine **2025**

Tech Stack: *CatBoost, Scikit-learn, Flask RESTful API, Python*

- Developed a predictive pricing model for food commodities utilizing CatBoost gradient boosting, achieving an outstanding 90.8% R^2 score on test datasets.
- Wrapped the trained model into a production-ready RESTful Flask API, enabling real-time pricing queries and integration with frontend consumer web applications.

CERTIFICATIONS & EDUCATION

AWS Certified AI Practitioner	*– Amazon Web Services*	**Completed**
Generative AI with Large Language Models	*– DeepLearning.AI / AWS*	**Completed**
Natural Language Processing Specialization	*– DeepLearning.AI*	**Completed**
Deep Learning Specialization	*– DeepLearning.AI*	**Completed**
Machine Learning Specialization	*– Stanford University / DeepLearning.AI*	**Completed**
Advanced Data Analytics Professional Certificate	*– Google*	**Completed**
Bachelor's Degree in Computer Science (Faculty of Computers & Information)	*– Mansoura University*	**2025 – 2029 (Expected)**